



Homeward

Discharge instructions patients can **actually** **follow**

Plain language, the patient's own language, every sentence checked against the chart, and a human on every risky line.

Open model on AMD Instinct MI300X · vLLM · ROCm

Verified

Siga tomando metoprolol succinate 50 mg una vez al día.

Matches fact F7 · back-translation agrees

Blocked

No beba más de 15 litros de líquido al día.

Source says 1.5 L · number changed in translation

Uncertain

Stop taking ibuprofen.

Source says pause until kidneys are rechecked

THE PATIENT

Maria, 72, going home after heart failure

Five medicine changes in one day: one new, one doubled, one paused, one stopped, one swapped with a 36-hour gap.

The instructions are in English, written for clinicians. Maria reads Spanish. Her daughter will manage her pills.

If Maria keeps taking ibuprofen, or drinks 15 litres instead of 1.5, she is back in hospital within the week.

Turn discharge instructions into a packet the patient can follow

Homeward rewrites the clinician's text in plain language and in the patient's own language, checks every sentence against the source, and sends only the risky ones to a human. It never diagnoses or adds medical advice.

Discharge text

VIETNAMESE

FRENCH

SPANISH

Homeward

Discharge instructions patients can follow

SPANISH

Heart failure, new and changed medicines, Spanish

Scripted sample run

ITALIAN

Knee replacement, opioid and blood thinner, Italian

Scripted sample run

Clinician's discharge instructions

DISCHARGE INSTRUCTIONS - Riverside General Hospital, Cardiology Unit

Patient: Maria Lopez MRN: 00482913 DOB: 03/14/1953

Attending: Dr. Anil Patel

Discharge date: 10/14/2026

Principal diagnosis: Acute decompensated heart failure with reduced ejection fraction (EF 30%), precipitated by dietary sodium indiscretion and NSAID use.

Hospital course: Patient admitted with dyspnea, orthopnea and bilateral lower-extremity edema. Diuresed with IV furosemide, net negative 4.2 L. Creatinine rose transiently to 1.6 mg/dL and improved to 1.3 mg/dL at discharge.

MEDICATIONS

Patient's language

Spanish

Patient name (masked before any model call)

Maria Lopez

Or upload a .txt or .pdf

Choose File

No file chosen

Create packet

Use synthetic or de-identified text only. This is a demo, not a medical device.

What happens

Mask identifiers

1 · Prepare

2 · Review

3 · Patient

4 · Impact

Sample runs only · no GPU connected

Send the text to the model

The model lists every instruction, each with an exact quote from the source.

Write in plain language

Grade 6 target. Every sentence cites the facts it restates.

Translate

Drug names, numbers and contact details kept as written.

Back-translate

A second pass turns the translation back into English for checking.

Safety model review

The model checks each sentence against its cited facts.

Deterministic checks

Numbers, drug names, stop vs keep, coverage of every critical fact.

Ready for human review

16 sentences, 6 need a human, 1 missing items.

Open review

Fact ledger

every instruction the model found, each tied to a quote from the source

ID	KIND	INSTRUCTION	RISK	CHECK
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THE EVIDENCE

Discharge is where instructions get lost

48%

of patients knew what
their medicines were for

Karliner et al., Medical Care 2012

50%

of limited-English
patients knew their
follow-up (vs 66%)

Karliner et al., Medical Care 2012

Grade 10

reading level of discharge
notes; the AMA
recommends grade 6

Mayo Clinic, Am J Surg 2016

–59%

post-discharge calls after
rewriting notes more
simply (22 → 9 per 100)

Mayo Clinic, ACS 2017

WHY AI ALONE IS NOT THE ANSWER

Rewriting is easy. Trusting the rewrite is the problem.

18%

of physician reviews of GPT-4 patient-friendly summaries raised a safety concern

Zaretsky et al., JAMA Netw Open 2024

24%

found omissions; 4% found hallucinations

Zaretsky et al., JAMA Netw Open 2024

"keep taking"

is what machine translation made of "hold the kidney medicine" in Chinese

Khoong et al., JAMA Intern Med 2019

HOW HOMEWARD WORKS

Every sentence earns its place

1 • Mask

Names, record numbers, birth dates, phones replaced before any model call

2 • Fact ledger

Every instruction tied to an exact quote; medicine list cross-checked by rules

3 • Plain draft

Grade 6, each sentence cites its facts

4 • Translate

...and back-translate to check meaning

5 • Check

Numbers, medicines, stop vs keep, contacts, coverage; safety model verdict

6 • Human

Only flagged and high-risk lines; named sign-off

7 • Teach-back

Quiz from approved sentences; wrong answers alert the nurse

Five schema-constrained model calls per packet, all on one AMD Instinct MI300X. Deterministic checks around the model, so its mistakes are caught instead of trusted.

Six rules, built into the workflow

REQUIREMENT	HOW IT IS HANDLED
No diagnosis or advice	Restates the clinician only. No cited fact or an unknown medicine means blocked.
Wrong or unsupported output	Quote check, rule-based medicine cross-check, number / medicine / stop-vs-keep checks, safety model.
Sensitive information	Identifiers masked before any model call; self-hosted model; memory-only cases.
Uncertain results	green amber red on every sentence, with the reason. Unsupported language or too many failures go to an interpreter.
High-risk decisions	Blood thinners, insulin, opioids, any stop or dose change, warning signs: always signed off by a person.
Human review	Nothing reaches the patient without a named sign-off. Red lines cannot be approved as they are. Full audit trail.

THE PRODUCT

Review in minutes, not a full re-read

Homeward

Discharge instructions patients can follow

1 · Prepare

2 · Review

3 · Patient

4 · Impact

Sample runs only · no GPU connected

16

1

1

5

1

10

sentences

blocked

uncertain

high risk

missing

verified, no review needed

Reviewer

RN Sam Lee

Show only what needs a human

Sign off and release

Still open: 7 item(s). S4 needs a decision. S5 needs a decision. S6 needs a decision. ...

Missing from the packet · F9

Not explained to the patient: Call the heart failure clinic at [PHONE_1] if your weight goes up more than 2 lb in one day or 5 lb in one week, or if your leg swelling gets worse.

Source: "Call the heart failure clinic at [PHONE_1] if weight increases by more than 2 lb in one day or 5 lb in one week, or if leg swelling worsens."

English

Spanish

Call the heart failure clinic at [PHONE_1] if your weight goes up more than 2 lb in one day or 5 lb in one week, or if your leg swelling gets worse

Suggest wording with the model

Add to packet

Not needed for this patient...

Verified

High risk

S4

ENGLISH

SPANISH

Your furosemide dose is now 40 mg. Take it 2 times a day, at 8 AM and 2 PM.

Su dosis de furosemide ahora es 40 mg. Tómela 2 veces al día, a las 8 de la mañana y a las 2 de la tarde.

Back-translation: Your dose of furosemide now is 40 mg. Take it 2 times a day, at 8 in the morning and at 2 in the afternoon.

! High-risk instruction: a person must confirm it even though every check passed.

✓ 3 check(s) passed

F3 · medication

Approve

Edit

Remove

Uncertain

High risk

S5

ENGLISH

SPANISH

Stop taking ibuprofen.

Deje de tomar ibuprofen.

Back-translation: Stop taking ibuprofen.

! Source says pause Ibuprofen until told to restart; this may read as stopping for good.

! Safety model: partly supported. Source says hold until kidney recheck; sentence reads as a permanent stop.

✓ 2 check(s) passed

F4 · medication

Approve

Edit

Remove

Sus instrucciones para volver a casa

Spanish

Both

English

Print

✓ Checked by RN Sam Lee

🏥

Por qué estuvo en el hospital

Leer en voz alta

Usted estuvo en el hospital porque su corazón no bombeaba bien y se acumuló líquido.

Demasiada sal y un medicamento para el dolor llamado AINE lo empeoraron.

💊

Sus medicamentos

Leer en voz alta

Empiece a tomar spironolactone 25 mg. Tómelo una vez al día.

Su dosis de furosemide ahora es 40 mg. Tómela 2 veces al día, a las 8 de la mañana y a las 2 de la tarde.

No tome ibuprofen hasta que su cardiólogo vuelva a revisar sus riñones y le diga que puede tomarlo.

Deje de tomar lisinopril. No lo vuelva a tomar.

Empiece a tomar sacubitril/valsartan 24/26 mg, 2 veces al día.

Espere al menos 36 horas después de su última dosis de lisinopril antes de su primera dosis de sacubitril/valsartan.

Siga tomando metoprolol succinate 50 mg una vez al día, igual que antes.

🚨

Pida ayuda si...

Leer en voz alta

Llame al 911 o vaya a la sala de emergencias si tiene dolor de pecho, dificultad para respirar en reposo o se desmaya.

Llame a la clínica del corazón al (555) 201-3344 si su peso sube más de 2 libras en un día o 5 libras en una semana, o si sus piernas se hinchan más.

📅

Sus citas

Leer en voz alta

Vea a Dr. Anil Patel en la clínica de insuficiencia cardíaca el 10/21/2026 a las 10:30 de la mañana.

Hágase un análisis de sangre en el laboratorio ambulatorio el 10/18/2026.

🏠

Cómo cuidarse en casa

Leer en voz alta

7

RESULTS

What it caught

8.7 → 3.8

reading grade, heart
failure case

Flesch-Kincaid on the English draft

63%

of sentences verified with
no human needed

the reviewer reads 6 of 16

3 / 3

planted problems
stopped: 15 L vs 1.5 L,
hold-as-stop, missing
warning

plus an invented ibuprofen in the
Italian case

99.4%

of 161 injected errors
flagged by deterministic
checks alone

0 false alarms on clean packets;
synthetic test

AMD

A 72B multilingual model on one GPU, inside the hospital

- vLLM + ROCm on an AMD Instinct MI300X (192 GB HBM3), AMD Developer Cloud
- 192 GB fits Qwen2.5-72B-Instruct in bf16 on a single GPU: no third-party API sees patient text
- vLLM batches a whole ward's discharges on one card
- Measured on MI300X: latency, tokens/s and GPU cost per packet (from `scripts/benchmark.py`)

Fewer calls back, fewer readmissions, less nurse time

- Every discharge, every language, at GPU cost of cents per packet
- Reviewer time goes to the 4 in 10 sentences that need it
- Teach-back catches misunderstanding before the patient leaves, not after
- Next: FHIR read from the EHR, more languages, pilot on one ward with nurse time and call-back rates measured

Homeward never diagnoses. It makes sure what the clinician wrote reaches the patient, in their language, and that they understood it.